SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Artificial Intelligence and Machine Learning

ASSESSMENT 3 – Technical Presentation(5 QUESTIONS)

Course Code:	ITA05	**Course Title:**	**Computer Vision**
CO Assessed:	CO1	**Assessment:**	Assessment 3: Technical Presentation
Weightage:	35% of CIA	**Total Marks:**	(5 Questions × 10 Marks)
CO1 Statement:	***Apply the fundamental concepts, image formation models, and image processing techniques to analyze and solve real-world Computer Vision problems.***		
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Section / Batch:	BE-CSE (Final Year)	**Date:**	14/10/2026

Code of Conduct

I Maredi Neela Meghana - 192311002 (NAME/ REG NO) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: M.NEELA MEGHANA

QUESTIONS: 5 **Total Marks: 50**

Annexure A

Technical Presentation

Q1. A shopping mall uses CCTV cameras that produce clear images during the day but dark and noisy images at night, making it difficult to identify people and objects.

Prepare a technical presentation demonstrating how filtering, noise removal, contrast enhancement, and sharpening techniques can be applied to improve the captured images. Present the sequence of image-processing operations required before object detection.

Problem understanding

A shopping mall's CCTV system captures clear images during the day but produces dark and noisy images at night. Low illumination reduces the visibility of people, clothing, faces, and objects. Sensor noise becomes more noticeable when the camera increases its sensitivity. If these images are sent directly to an object-detection system, the detector may miss objects or produce false detections. The aim is therefore to improve image quality while preserving useful details before object detection.

Theory and processing stages

1. Image acquisition and inspection. Read the CCTV frame and examine its brightness, noise level, and important regions. Keep the original frame unchanged so the enhanced result can be compared with it.
2. Noise removal using filtering. Apply a median filter to reduce salt-and-pepper noise while preserving edges. A Gaussian filter can reduce fine random noise, but a large kernel may blur faces and object boundaries. Select the filter and kernel size carefully.
3. Contrast enhancement. Use histogram equalization to spread intensity values across the available range. For local enhancement, apply CLAHE (Contrast Limited Adaptive Histogram Equalization), which improves dark regions while limiting excessive amplification of noise.
4. Brightness correction. If the whole frame is underexposed, apply a controlled gamma correction with gamma below 1. Avoid making bright lamps or reflective surfaces completely white.
5. Sharpening. Apply an unsharp mask or a mild Laplacian-based sharpening step after denoising. Sharpening increases the visibility of boundaries, but excessive sharpening can create halos and amplify remaining noise.
6. Object detection. Pass the processed image to the detector. Evaluate detections against the original image and monitor false positives, missed objects, and confidence scores.

Technique comparison table

Technique	Main purpose	Strength	Limitation
Median filtering	**Remove impulse noise**	**Preserves many edges**	**May remove tiny details**
Gaussian filtering	**Reduce random fine noise**	**Smooth and simple**	**Can blur boundaries**
Histogram equalization	**Improve overall contrast**	**Simple global improvement**	**May over-enhance some areas**
CLAHE	**Improve local contrast**	**Works well in uneven lighting**	**Can reveal noise if overused**
Gamma correction	**Brighten dark regions**	**Direct brightness control**	**Can wash out highlights**
Unsharp masking	**Make edges clearer**	**Improves visible detail**	**Can amplify noise and halos**

Recommended pipeline

Recommended order: inspect image → denoise → correct brightness/contrast → sharpen gently → detect objects → evaluate results. The exact order can be adjusted after testing; denoising before sharpening is especially important because sharpening noise would make the image worse.

Conclusion

Quality can be checked using side-by-side visual comparison, brightness distribution, edge clarity, and object-detection precision and recall. A good result is not simply the brightest image: it must show useful details without creating artificial edges. Night-time CCTV enhancement can improve monitoring, but it cannot recover information that was never captured because of severe darkness, motion blur, or obstruction.

Q2. A university is developing an automated document-scanning system using a camera, where captured documents may contain shadows, noise, uneven illumination, tilted pages, and blurred text.

Prepare a technical presentation explaining how filtering, thresholding, contrast enhancement, edge detection, and segmentation can be applied to improve the quality of the document images. Demonstrate the role of each technique in the processing pipeline.

Problem understanding

A camera-based document scanner must convert a photographed page into a clear, readable digital document. Shadows, uneven illumination, noise, tilted pages, and blurred text can reduce OCR (Optical Character Recognition) accuracy. The processing pipeline should correct the page appearance, separate text from the background, identify page boundaries, and produce a clean image for reading or OCR.

Theory and processing stages

1. Image acquisition and page correction. Capture the full page with as little motion as possible. Detect the page boundary and correct perspective using a geometric transform. Deskew the page so text lines become approximately horizontal.
2. Filtering and noise removal. Use a small median filter for isolated speckles or a Gaussian filter for mild sensor noise. Use conservative settings because strong smoothing can erase thin characters, punctuation, and small print.
3. Illumination and contrast enhancement. Estimate and reduce slow changes in background brightness. Contrast stretching or CLAHE can improve faint text. Background normalization is useful when one side of the page is darker than the other.
4. Thresholding. Convert the grayscale image to foreground text and background paper. Global thresholding works when illumination is uniform. Adaptive thresholding calculates thresholds for local regions and is usually better when shadows or lighting vary across the page.
5. Edge detection. Apply an edge detector such as Canny to help locate the page outline or document borders. Edge detection is a supporting step; it does not itself remove shadows or classify all text pixels correctly.
6. Segmentation and output. Separate the page, text blocks, lines, and sometimes individual characters. Remove small isolated components only when they are likely noise. Save the corrected image and send it to OCR.

Technique comparison table

Technique	Role in scanner	Best use case	Caution
Median/Gaussian filter	**Reduce image noise**	**Speckles or mild sensor noise**	**Over-filtering removes thin strokes**
Contrast enhancement	**Separate faint text from paper**	**Low-contrast printing**	**May amplify paper texture**
Global thresholding	**One threshold for the full image**	**Even lighting and plain paper**	**Fails under strong shadows**
Adaptive thresholding	**Local threshold for image regions**	**Uneven illumination**	**Small windows may create patchy text**
Canny edge detection	**Find page boundaries and edges**	**Page outline detection**	**Text and background can create extra edges**
Segmentation	**Separate page/text regions**	**OCR preparation**	**Touching characters may merge**

Recommended pipeline

Suggested pipeline: capture → page-boundary detection and perspective correction → denoising → background/contrast correction → grayscale thresholding → edge-assisted page refinement → text segmentation → OCR and quality check.

Conclusion

Test the result using readability, character recognition accuracy, missing-stroke rate, and processing time. Keep a grayscale version when a binary image removes faint details. A practical scanner may use adaptive thresholding for the final text image while retaining the enhanced grayscale image as a backup. No single technique works best for every page, so the pipeline should adapt to lighting and document quality.

Q3. A fruit-processing company wants to automatically identify damaged apples from images captured on a conveyor belt, where shadows, reflections, different backgrounds, and brightness variations are present.

Prepare a technical presentation comparing global thresholding, adaptive thresholding, edge-based segmentation, and region-based segmentation for this application. Demonstrate how a suitable technique or combination of techniques can improve fruit defect detection.

Problem understanding

A fruit-processing company wants to detect damaged apples on a conveyor belt. Shadows, reflections, background changes, and brightness variation make simple colour or intensity rules unreliable. Segmentation separates the apple from the background or identifies suspicious regions on its surface. The segmented regions can then be measured or passed to a classifier to decide whether an apple is damaged.

Theory and processing stages

1. Image preparation. Capture images with a consistent camera position where possible. Apply light denoising and correct major brightness variation. If available, use diffuse lighting to reduce hard shadows and reflections.
2. Global thresholding. Choose one intensity or colour threshold for the entire image. This is fast and simple when the apples and background have clearly different values, but it is sensitive to shadows and changing illumination.
3. Adaptive thresholding. Calculate a threshold from each local neighbourhood. This can handle uneven lighting better than a single global threshold, but reflections or texture may create fragmented masks.
4. Edge-based segmentation. Use an edge detector to identify boundaries between the apple and background, or between healthy and visibly damaged areas. It is useful when boundaries are clear, but weak, broken, or noisy edges can cause incomplete regions.
5. Region-based segmentation. Group neighbouring pixels with similar colour, intensity, or texture. Region growing or clustering can separate apple regions and surface patches, but results depend on seed selection or similarity settings.
6. Defect decision. After segmenting each apple, extract features such as defect area, colour difference, texture, shape, and location. Use thresholds or a trained classifier to label the apple as normal, damaged, or requiring inspection.

Technique comparison table

Method	How it works	Advantages	Weakness in this case
Global thresholding	Uses one threshold for the full image	Fast; easy to implement	Shadows and reflections change pixel values
Adaptive thresholding	Uses local thresholds	Handles uneven illumination better	Can produce noisy or fragmented masks
Edge-based segmentation	Finds intensity boundaries	Useful for clear outlines and cracks	Broken edges can leave gaps
Region-based segmentation	Groups similar neighbouring pixels	Can use colour and texture together	May merge similar apple/background regions
Combined approach	Preprocesses, segments, then validates regions	More robust to varied conditions	Requires tuning and testing

Recommended pipeline

Recommended approach: denoise and normalize lighting → segment the whole apple using colour/region information → refine its boundary with edges → inspect the apple surface for abnormal colour or texture → extract defect features → classify and flag uncertain cases.

Conclusion

Global thresholding is suitable for a controlled line with stable lighting. Adaptive thresholding is preferable when illumination changes locally. Edge-based methods are valuable for clear cracks or boundaries, while region-based methods can use colour and texture to group defect areas. A combined approach is generally more reliable than relying on one technique alone. The system should be tested on apples with different varieties, orientations, levels of damage, and background conditions. Human review should be available for uncertain predictions.

Q4. An autonomous vehicle uses a front-facing camera to detect lane markings under different weather and lighting conditions, where shadows, damaged markings, vehicles, and rain may affect the captured images. **Prepare a technical presentation demonstrating how grayscale conversion, filtering, contrast enhancement, edge detection, thresholding, and region-of-interest techniques can be applied for lane detection. Explain the challenges associated with each processing stage.**

Problem understanding

A vehicle's front-facing camera must identify lane markings so the driving system can estimate lane position and road direction. Shadows, rain, worn paint, glare, road texture, and nearby vehicles can confuse image-processing methods. A practical pipeline reduces irrelevant variation, highlights likely lane markings, and restricts analysis to the road region. The result must be checked over consecutive frames because a single frame can be misleading.

Theory and processing stages

1. Grayscale conversion. Convert the colour image to grayscale to simplify processing and reduce the data used by later intensity-based operations. Some lane markings are distinguished by colour, so colour information may also be retained as a parallel cue.
2. Filtering. Use a Gaussian filter to reduce small texture variations and sensor noise. Excessive smoothing can erase thin or damaged lane markings, so the filter size should remain small.
3. Contrast enhancement. Improve the visibility of faint lane paint using contrast stretching or local enhancement. Avoid over-enhancing road texture, glare, or rain reflections, which may become false lane candidates.
4. Edge detection. Use Canny edge detection to locate strong intensity transitions that may correspond to lane boundaries. Shadows, cracks, curbs, and vehicle edges can also be detected, so edges alone are insufficient.
5. Thresholding. Apply intensity or colour thresholds to retain likely bright or yellow lane markings. Thresholds may need adaptation to daylight, tunnels, wet roads, and night-time scenes.
6. Region of interest (ROI). Mask the image to focus on the road area where lane markings are expected. This reduces irrelevant detections from buildings, trees, and the sky, but an incorrectly chosen ROI can exclude curved roads or nearby lane segments.

7. Lane estimation and temporal checking. Group candidate pixels or lines and estimate lane curves. Compare detections across consecutive frames to reduce sudden jumps. If confidence is low, the vehicle should rely on

other sensors and safe fallback behaviour.

Technique comparison table

Stage	Purpose	Typical challenge	Possible response
Grayscale conversion	Simplify intensity analysis	Colour cues may be lost	Retain colour cues when useful
Filtering	Reduce noise and road texture	Thin markings may disappear	Use mild smoothing
Contrast enhancement	Reveal faded paint	Glare and texture may be amplified	Limit local enhancement
Edge detection	Find marking boundaries	Cracks and shadows create edges	Combine with thresholding and ROI
Thresholding	Select likely lane pixels	Lighting changes alter pixel values	Use adaptive or colour-aware rules
ROI selection	Ignore irrelevant image areas	Curves and lane changes may fall outside ROI	Use a flexible road-region mask
Temporal checking	Stabilize lane estimates	Rapid changes or occlusion	Track confidence and allow fallback

Recommended pipeline

Pipeline: camera frame → grayscale/colour cues → filtering → contrast enhancement → edge detection and thresholding → road ROI → lane grouping/curve fitting → temporal validation → lane estimate with confidence.

Conclusion

Lane detection is safety-critical, so it should not depend on a single threshold or one frame. Performance must be tested in rain, shadows, worn markings, glare, night scenes, curves, and partial obstruction by vehicles. The

system should report low confidence when markings cannot be identified reliably rather than inventing a lane position. In production vehicles, camera-based processing is combined with additional sensing, tracking, and safety logic.

Q5. A municipal corporation wants to classify waste such as plastic, paper, and metal using a camera installed above a conveyor belt. The captured images may contain overlapping objects, shadows, reflections, noise, and different object orientations.

Prepare a technical presentation demonstrating a complete image-processing pipeline using preprocessing, enhancement, segmentation, feature extraction, and classification. Explain how the proposed pipeline can be applied to handle real-world variations and improve waste classification.

Problem understanding

A municipal corporation wants to classify plastic, paper, and metal waste from a camera mounted above a conveyor belt. Real images may contain overlapping objects, shadows, reflections, noise, and different orientations. A complete pipeline should improve image quality, separate objects where possible, extract useful visual features, and classify each item. Because some objects look similar, the system should also identify uncertain cases for manual sorting.

Theory and processing stages

1. Image acquisition and preprocessing. Capture images with a fixed camera and adequate lighting. Resize consistently, correct colour if needed, and apply mild median or Gaussian filtering to reduce noise without erasing texture.
2. Image enhancement. Use contrast adjustment or CLAHE to make boundaries and surface details easier to see. Correct uneven illumination where possible. Avoid excessive enhancement because shiny metal and transparent plastic can create misleading highlights.
3. Segmentation. Separate foreground waste from the conveyor background using colour, thresholding, background subtraction, edges, or a segmentation model. For overlapping objects, use connected components and shape cues; touching items may need watershed-style separation or a learned segmentation model.
4. Feature extraction. Measure shape, area, aspect ratio, contour, colour, texture, and reflective appearance. Traditional features can support a simple classifier. A convolutional neural network can learn more complex patterns directly from image data.
5. Classification. Classify each segmented item as plastic, paper, or metal using a trained model such as a support vector machine for engineered features or a CNN for image features. The training dataset should include varied lighting, orientation, object types, and examples of dirty or crushed materials.

6. Post-processing and sorting. Check confidence scores, remove duplicate detections, and track items as they move along the belt. Send low-confidence or heavily overlapping objects for a second view or human review. Use the class output to control a sorting mechanism only after safety checks.

Technique comparison table

Pipeline stage	Methods/examples	Benefit	Real-world issue handled
Preprocessing	**Resize, denoise, colour correction**	**Consistent input**	**Sensor noise and size variation**
Enhancement	**Contrast stretching, CLAHE**	**Improves visible details**	**Shadows and uneven lighting**
Segmentation	**Thresholding, background subtraction, edges, watershed**	**Separates objects from background**	**Different backgrounds and touching items**
Feature extraction	**Shape, colour, texture, reflectance**	**Provides evidence for classification**	**Different orientations and surface patterns**
Classification	**SVM or CNN**	**Assigns material class**	**Variation among plastic, paper, and metal**
Post-processing	**Confidence threshold, tracking, review**	**Reduces duplicate or unreliable results**	**Overlap and uncertain predictions**

Recommended pipeline

Proposed pipeline: capture image → preprocess → enhance → segment waste objects → separate touching objects → extract features → classify as plastic/paper/metal → check confidence → track and sort or request manual review.

Conclusion

Reliable classification depends on the quality and diversity of training data, not only on the algorithm. The dataset should include transparent and coloured plastic, printed and crumpled paper, cans and other metal items, dirt, partial occlusion, and different orientations. Evaluate the system using per-class precision, recall, confusion

matrix, and overall accuracy. A confidence threshold and human-review route help prevent incorrect sorting. The system should be tested on a real conveyor before automatic mechanical sorting is enabled.

Rubrics for Calculation

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Technical Content Accuracy	30	Highly accurate, indepth, and insightful technical explanation	Technically correct content with adequate depth and clarity	Basic concepts presented with limited accuracy and depth	Content contains major technical errors; concepts poorly understood
Problem Identification	25	Problem rigorously analyzed using appropriate and advanced methods	Problem clearly defined with a logical engineering approach	Problem identified but analysis or methodology is weak	Problem statement unclear or incorrect; no logical approach
Use of Tools	20	Advanced tools, well-analyzed data, and highly effective visuals	Appropriate tools and data used effectively with clear visuals	Limited use of tools and basic data interpretation	Minimal or incorrect use of tools and data; poor visuals
Communication	15	Highly engaging, confident, and audience-focused delivery	Clear, organized, and professional communication	Understandable but lacks clarity, structure, or confidence	Poor organization, unclear speech, ineffective slides
Professionalism	10	Exemplary professionalism, leadership, and ethical awareness	Professional conduct with effective team collaboration	Basic professionalism; uneven team participation	Lacks professionalism; minimal contribution or ethical awareness

Faculty In-charge	Course Coordinator	Program Director
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____